higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

9 April 2025

This marking guideline consists of 6 pages.

Copyright reserved

GET IT ON Google Play

Try our Apps on Play Store

NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	More…
N4-N6 Computer Practice NATED	**N4-N6** Entrepreneurship and Business Management NATED	**N4** Management Communication NATED	**N4-N6** Communication NATED	**N4-N6** Public Administration NATED	**N4-N6** Financial Accounting NATED
N4-N6 Computerised Financial Systems NATED	**N5-N6** Sales Management NATED	**N4-N6** Office Practice NATED	**N5-N6** Public Finance NATED	**N4-N6** Personnel Management NATED	**N5-N6** Municipal Administration NATED
N4-N6 Information Processing NATED	**N5-N6** Personnel Training NATED	**N4-N6** Travel Services NATED	**N6** Public Law NATED	**N5-N6** Cost and Management Accounting NATED	More…

QUESTION 1

1.1 1.1.1 B
1.1.2 B
1.1.3 D
1.1.4 B
1.1.5 B

(5 × 1) (5)

1.2 1.2.1 False
1.2.2 True
1.2.3 False
1.2.4 True
1.2.5 True

(5 × 1) (5)
[10]

QUESTION 2

2.1 2.1.1 $11,25_{10} = 1011,01_2$ ✓ & $16,4_8 = 1110,100_2$ ✓

Two's complement of 1110,100 = 0001,011 + 1 = 0001,100 ✓

	0	*	1	0	1	1,	0	1	0	
+	1	*	0	0	0	1,	1	0	0	
	1	*	1	1	0	0,	1	1	0	✓

Two's complement of 1* 1100,110 = -0011,001 + 1 = $-11,010_2$ ✓

$-11,010_2$ - 3, 25_{10} ✓

2.1.2 $26,3_8$ ¿ $10110,011_2$ ✓ & 9, $5_{10} = 1001,1_2$ ✓

						1	**0,**	**0**	**1**	**0**	✓	
1	0	0	1	1	1	0	1	1	0	0,	1	1	0	
					1	0	0	1	1	↓	↓	↓	↓	✓
					0	0	0	1	1	0	1	1	↓	
								1	0	0	1	1	↓	
								0	1	0	0	0	0	✓

$10,010_2 = 2,2_8$ ✓

2.1.3 13, 375_{10} = $1101,011_2$ ✓ & 3, 2_8 = 011, 010_2 ✓

			1	1	0	1	0	1	1	
			X			1	1	0	1	
			1	1	0	1	0	1	1	✓
		0	0	0	0	0	0	0		
	1	1	0	1	0	1	1		✓	
1	1	0	1	0	1	1				
1	0	1	0	1	1	0	1	1	1	✓

10101101111= $101011,01111_2$ =$2B,78_{16}$✓

(3 × 6) (18)

2.2 2.2.1

8	4,	5	
+3	+3	+3	
11	7	8	✓
↓	↓	↓	
1011	**0111,**	**1000**	✓

84, 5_{10} = $10110111,1000_{xs3}$ code✓

2.2.2

8	4,	5	
2421	2421	2421	
↓	↓	↓	
1110	**0100,**	**1011**	✓✓

84, 5_{10} = 11100100, 1011_{2421} code✓

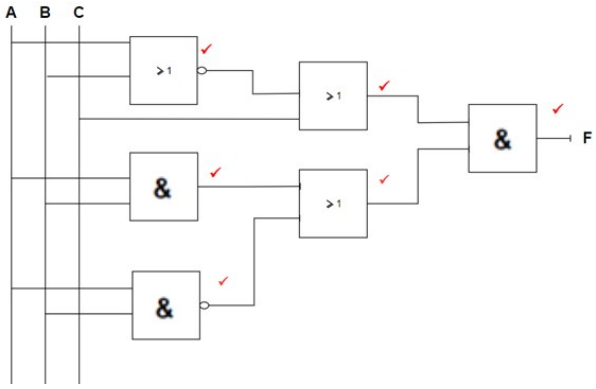
(2 × 3) (6)
[24]

QUESTION 3

3.1 3.1.1 $Q=\overline{AB}$
3.1.2 $Q=A+B$
3.1.3 $Q=\overline{A}\,B \vee Q=\overline{A}\,\overline{B}+AB$
3.1.4 $Q=\overline{A}$
3.1.5 $Q=AB$

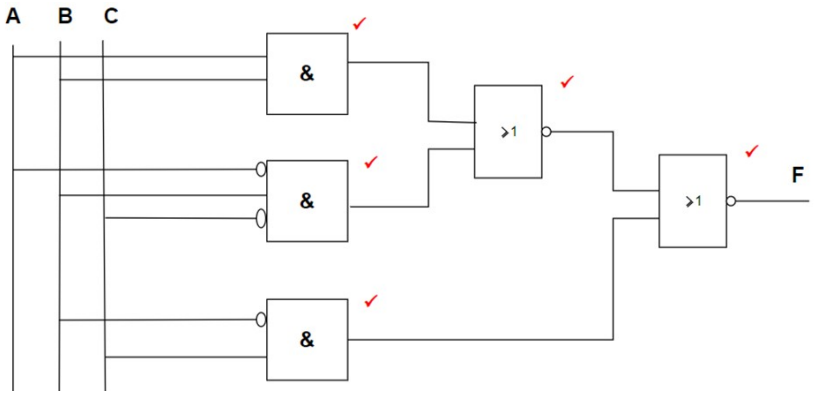
(5 × 1) (5)

3.2 3.2.1 $\overline{A+B}+C)(AB+\overline{AB})=F$



(6)

3.2.2 $AB+\overline{A}B\overline{C}+\overline{B}C=F$



(5)

3.3 3.3.1 $Z(\overline{X}+\overline{Y})+\overline{X}(Y+\overline{Z})+\overline{X}Y=F$

$\overline{X}(Z+\overline{Z})+\overline{Y}Z+\overline{X}Y$ ✓✓
$\overline{X}(1)+\overline{Y}Z+\overline{X}Y$ ✓
$\overline{X}+\overline{Y}Z+\overline{X}Y$ ✓
$\overline{X}+\overline{Y}Z$ ✓

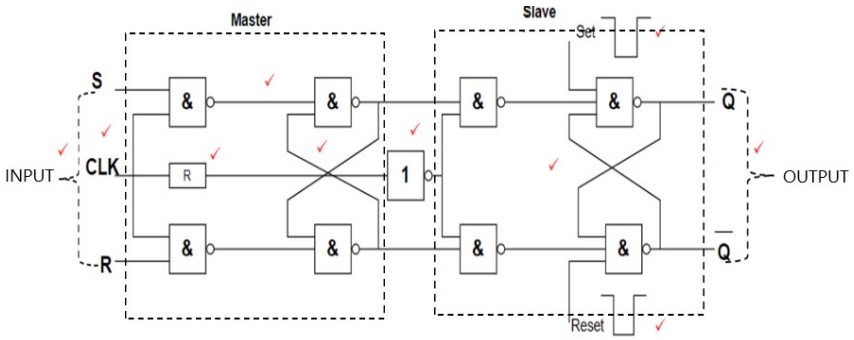
3.3.2 $\overline{A}BC+A\overline{B}C+\overline{A}B\overline{C}+\overline{A}\overline{B}C=F$

$\overline{A}B(C+\overline{C})+\overline{B}C(A+\overline{A})$ ✓✓
$\overline{A}B(1)+\overline{B}C(1)$
$\overline{A}B+\overline{B}C$ ✓✓

(2 × 5) (10)
[26]

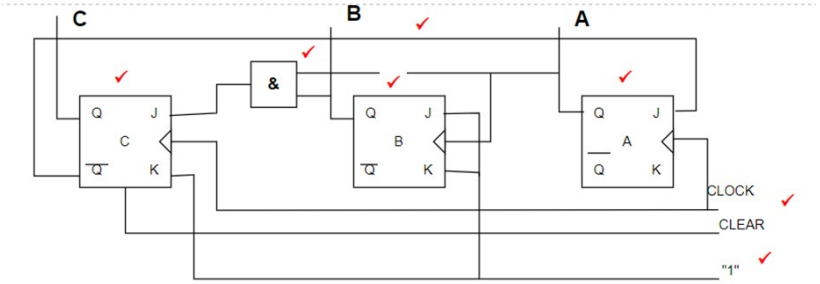
QUESTION 4

4.1 SR master slave flip-flop using NAND



(10)

4.2 Module 5 counter and its truth table

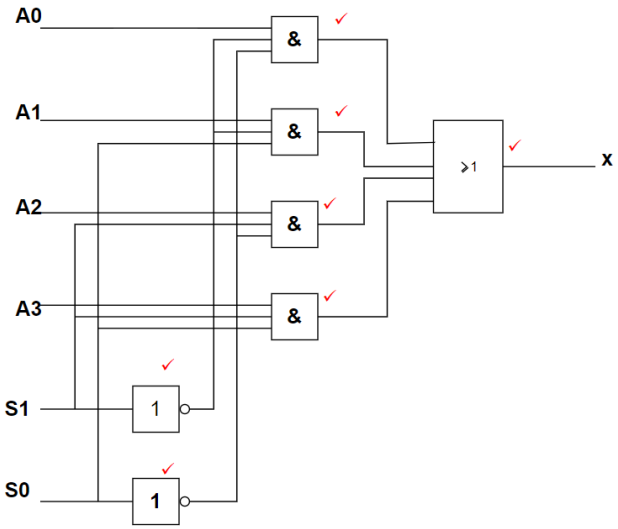


Pulse	C	B	A
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	0	0	0

(10)
[20]

QUESTION 5

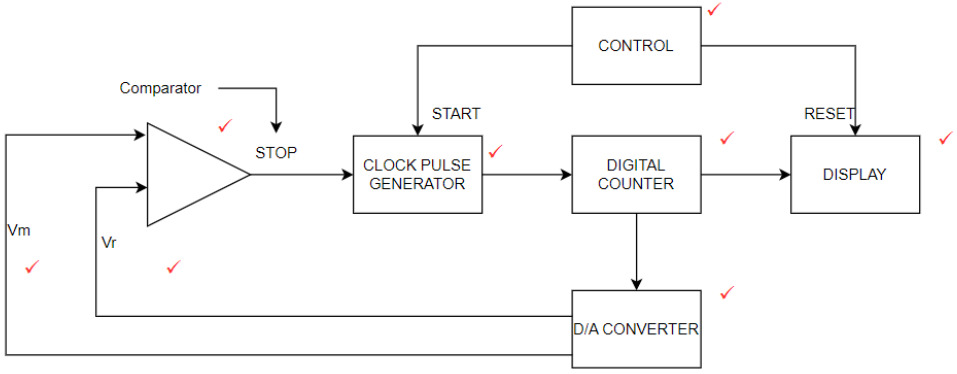
5.1 4 × 1 line multiplexer

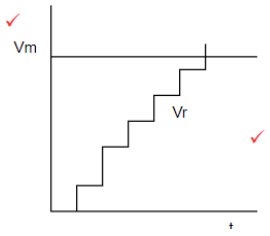


(7)

5.2 $V=\frac{1+2+4}{15}\times 24=11.2V$ (3)

5.3





(10)
[20]

TOTAL: 100

Copyright reserved